

FINAL PAPER

Surrogate-Aware Runtime Assurance for Agentic Medical LLMs

A synthetic stress-test framework for evidence-gated clinical claims

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Abstract

Medical large language models (LLMs) are often evaluated for hallucination and citation accuracy, but agentic clinical workflows introduce a subtler failure: evidence overclaiming by systems that retrieve, summarize, plan, and draft clinical-sounding actions. Building on prior course evaluations of LLMs, decision support, automation, and epidemic simulation, this paper proposes a runtime-assurance framework that gates LLM outputs and proposed actions by citation validity, endpoint strength, study design, target-population fit, actionability, opportunity cost, clinician authority, source provenance, runtime budget order, confounding, proportional claim exposure, witness-selection bias, privacy loss, and consent aggregation. A synthetic stress-test set distinguishes hard outcomes from surrogate/process endpoints, flags unverifiable citations, treats low clinician uptake as possible safety filtering, and requires abstention when context is missing. Additional probes cover multimodal telemedicine, consumer heart-rate/wearable physiology, cross-script and Romani-style language provenance, KAN-style architecture claims, SIK3/tinnitus genetics, saliva evolutionary genomics, altitude/circadian transport, robotic-cadaver rehearsal, neuropsychological norm drift, sequential depletion of scarce runtime resources, and family genomic consent under differential-privacy-style budget accounting. A model-selection audit then chooses the least unsafe claim form compatible with the available evidence. The framework is motivated by Medical Nihilism, AI Safety Gridworlds, Po-Ru Loh-style bioinformatics, differential privacy, social-choice limits, experimental mathematics, residual-depletion ordering, and pragmatic randomized-trial standards. It does not validate a clinical product, but provides a testable architecture for safer clinician-facing agentic medical LLMs.

1. Introduction

The central problem for medical LLMs is not simply whether they can produce plausible clinical text. It is whether agentic workflows that retrieve, summarize, plan, rank, and draft actions can keep the evidentiary strength of a claim separate from the fluency of the answer. A model may cite a real study and still make an unsafe inference if it treats screening completion, detection rate, evolutionary-genomics mechanism, or workflow efficiency as proof of improved mortality, morbidity, function, or quality of life.

This paper frames that problem as runtime assurance for bounded medical agency. The LLM is allowed to act as a bounded assistant, second reader, summarizer, retriever, or question generator only after a monitor has

classified the evidence and set a maximum permission level for the response or proposed action. In this framing, abstention is not a defect in the interface. Abstention is the behavior that prevents local evidence from being inflated into global clinical authority.

The Wald survivorship-bias aircraft diagram is a useful warning for medical LLM evaluation. If reviewers only inspect fluent responses that return with plausible citations, they may miss the more important denominator: prompts that should have abstained, evidence chains that never terminate in a patient outcome, populations excluded from validation, or contexts that disappeared during summarization. The missing marks are often where the safety system should reinforce itself.

2. Background

2.1 Medical Nihilism and Endpoint Discipline

Stegenga's Medical Nihilism argues for caution in medicine because biases, malleable study designs, and selective evidentiary practices can overstate benefits and understate harms. The point is not that medicine cannot work. The point is that the path from evidence to treatment confidence is fragile. Medical LLMs make this fragility operationally important because they can compress a messy literature into a confident answer that sounds more settled than the evidence itself.

FDA and FDA-NIH BEST materials provide the practical distinction needed for the runtime layer: clinical outcomes measure how people feel, function, or survive; surrogate endpoints can be useful but require validation and, in many cases, confirmatory evidence. This distinction is the backbone of the gate.

2.2 Pragmatic RCTs and Workflow Reality

Byrne, Domenico, and Moore argue that patient-outcome claims about healthcare AI should be tested in real workflows through pragmatic randomized trials. The CLOT trial illustrates why this matters. It embedded a validated prognostic model into clinical workflow, yet HA-VTE rates were not reduced and recommendations were accepted only 25.8 percent of the time. A shallow interpretation calls that last-mile failure. A safer interpretation asks whether clinicians were filtering recommendations for reasons the model did not encode.

This capstone therefore treats low acceptance, opportunity cost, and clinician concern as signals to be modeled rather than nuisances to be optimized away. The same logic applies to models such as ANA triage systems: accuracy can be valuable, but if the model diverts scarce specialist attention or hides uncertainty, it may not improve care.

2.3 Claim Grammar and Bounded Ambiguity

The formal-language analogy is deliberately bounded. Gold-style learnability asks what can be identified from a stream of positive examples. Carroll presents formal language theory and Gold learnability as tools for evaluating grammar systems. Kanazawa's work on k-valued categorial grammars gives the useful parameter: a symbol may receive at most k categories. In this paper, a citation, observation, or trial result is an evidence asset that may receive only a bounded set of evidence categories. The runtime monitor must resolve or preserve that ambiguity before an LLM composes a clinical claim.

Vishnikin and Okhotin's unique-category assignment result sharpens the intuition. Removing lexical ambiguity before composition can still leave expressive power. Analogously, a medical LLM does not need to erase nuance; it needs to prevent a single paper from being spent simultaneously as workflow evidence, surrogate evidence, and hard-outcome evidence.

The word elasticity is used here in Kanazawa's inductive-inference sense, not in the rubber-band sense of knot theory or Morse-theoretic visualization. A separate probability analogy comes from concentration and superconcentration: a runtime monitor should notice when variability, ambiguity, or endpoint roughness is being hidden by a smooth summary.

Van Vu's work in random graph theory and random matrix theory is included only as background for this stability intuition: high-dimensional AI systems often look smooth in aggregate while depending on spectral, concentration, and perturbation behavior that must be monitored rather than assumed away.

The bioinformatics anchor is closer to Po-Ru Loh-style statistical genetics than to an ordinary chatbot benchmark. Mixed models, linkage disequilibrium, admixture inference, and population structure all teach the same lesson: hidden structure can make a superficially portable signal confounded, miscalibrated, or population-specific. A medical LLM that reads genetic or polygenic-risk evidence should therefore expose ancestry/population fit, imputation status, uncertainty, and latent-factor handling before converting the evidence into a medical claim.

The clinical genetics scope is deliberately ordinary and hospital-facing. Hospitals repeatedly encounter medication-gene interactions such as CYP2C19 and clopidogrel, positive newborn-screening follow-up that points clinicians to ACT sheets and algorithms, reportable ACMG secondary findings, CDC Tier 1 applications such as hereditary breast and ovarian cancer, Lynch syndrome, and familial hypercholesterolemia, and variants of uncertain significance that must not be overinterpreted. These are monthly or yearly governance and decision-support problems in real systems. They are a better first test bed than speculative designer-baby, open-ended wellness, or unconstrained polygenic-risk claims.

Dwork and Roth provide the formal privacy backbone. Differential privacy is a definition rather than one algorithm; privacy loss is quantified by parameters such as epsilon and delta; composition consumes privacy budget; group privacy matters for families; and post-processing does not increase privacy loss if the underlying release was already private. For medical genetics, this means the access-control question is not simply who sees the full record. The runtime monitor must track which family-linked genomic coordinates, phenotype summaries, or polygenic-risk features are released, at what granularity, under what cumulative privacy budget.

Arrow's theorem adds a consent-aggregation warning. Ranked preferences from relatives over options such as full raw reuse, trait-specific partial inclusion, audit-only use, or no reuse cannot always be collapsed into one fair family ordering. In the proposed framework, ranked choice is useful input, but the runtime decision preserves individual vetoes, partial permissions, and audit logs. This is the graded-poset point: moving downward from a full family genomic release can happen along multiple incomparable paths, not along one universal chain.

Boltzmann fair division and softmax policy mirror ascent are used only inside that boundary. Entropy-controlled or softmax-style allocation can prioritize limited audit attention, model-routing probability, or privacy budget among feasible options, and SPMA supplies a modern mirror-ascent reference for fast softmax policy updates in bandit and tabular MDP settings. Neither method overrides consent or converts a governance decision into clinical evidence.

Uniform generation of NP-witnesses supplies a final evaluation analogy. An evidence witness is a concrete support pattern for a claim: endpoint, design, population, and provenance. A medical LLM evaluator that samples only the strongest-looking witnesses can overstate the system's reliability. The proposed audit therefore records witness-selection bias rather than treating cherry-picked examples as representative.

The measure-theory language is used in a finite, audit-friendly sense. An event is a measurable set of runtime histories, for example the set where a surrogate claim is promoted to a hard-outcome claim, privacy budget is exceeded, or a consent boundary is collapsed. A population, hospital, family, or evidence source can then be represented as a probability measure over those histories. A measure on the space of measures represents uncertainty over which source distribution is active. This is the abstract form of subgroup and data-source robustness: the mixture may look safe while one source measure remains high-risk.

Sheffield's lecture on zero-one laws and maximal inequalities supplies a compact probability lens for runtime assurance. A tail event is a long-run property whose membership does not change when finitely many early observations are altered. That is a warning against certifying an agentic medical LLM from a short clean prefix. Kolmogorov's maximal inequality adds a finite-horizon monitor: even when increments are mean-zero, the relevant safety question may be whether the cumulative process crossed a boundary at any time, not merely where it ended.

Karlin and Peres's Hex and Y exposition supplies a discrete topological analogy for terminal safety states and safe coarse-graining. Hex is finite and progressively bounded, and a completed standard board has exactly one winning crossing. Their Y-board reduction also uses local majority recoloring of small triples while preserving the global crossing/Y property. The capstone uses that idea as a design invariant: after required evidence features are known, a runtime state should not be both permitted and blocked, or neither classified; and any privacy or genetics coarse-graining should preserve the boundary property it claims to support. The strategy-stealing lesson is also useful for agentic systems: an unlogged extra move or evidence feature can advantage the acting agent, so hidden moves must be treated as safety-relevant.

Mahalanobis distance makes this operational for covariate balance. Instead of asking whether a source population seems similar by one variable at a time, the monitor measures covariance-adjusted distance from a reference vector of ancestry principal components, age, rural access, missingness, and privacy constraints. Large distance is a runtime transport warning: the source needs source-specific validation before its evidence is used as if it belonged to the reference distribution.

Latent Dirichlet Allocation adds a familiar machine-learning instance of this viewpoint. LDA represents documents as random mixtures over latent topics. In the capstone, an evidence artifact can analogously be represented as a mixture over latent claim topics such as hard outcome, surrogate, workflow, privacy, or consent. The runtime monitor can use such mixtures for routing and triage, but the mixture itself is not a clinical endpoint.

Figaro is a natural future substrate for this layer because it is a probabilistic programming language for expressing rich probabilistic models and applying reasoning algorithms to evidence. This submission keeps the executable artifact in dependency-light Python, but the same finite event algebra, measure-on-measures source model, and runtime-history gates could be recast in Figaro for richer probabilistic replay and audit.

Connes's use of Penrose tilings in noncommutative geometry supplies a second bounded analogy. The classical quotient of all Penrose tilings by translation is pathological because local patches recur everywhere while global tilings remain nonidentical; the remedy is to keep relational structure rather than replacing the space by a trivial quotient. Medical evidence has a similar failure mode for LLMs: the same finite patch of words, citations, biomarkers, or workflow metrics may recur across settings, but quotienting away population, time, instrument version, clinical role, and action context can erase the structure that makes a claim safe or unsafe.

Cohen's forcing story gives a third bounded logic analogy. In set theory, a statement may be independent of a chosen axiom system. In medical LLM evaluation, a patient-outcome claim may be independent of the evidence package currently available to the agent. The safe runtime response is to expose the missing assumption, request additional evidence, or abstain, rather than letting fluent language act as a new axiom.

The nonlocal, inpainting, and coding-theory references sharpen the same point from three directions. Du's nonlocal modeling emphasizes that some systems depend on interactions at a distance or across memory; family genetics, longitudinal records, and social-language context have that character. Carlini and Tozza's game p-Laplacian inpainting work is a useful caution for clinical AI: reconstructing missing parts can be principled, but reconstructed content must remain labeled as reconstruction rather than observed evidence. Korner's coding notes add the decodability requirement: after compression, captioning, blurring, encryption, or summarization, the downstream system must still know which evidence object it is reading.

Jaffe and Liu's mathematical picture-language program makes the multimodal version precise. They distinguish an abstract picture language L , a target reality R , and a simulation map S connecting the two. In medical AI, a screenshot, wound photo, imaging slice, waveform, captioned video, or diagram can be a powerful object in L . The runtime gate asks whether the simulation map to the clinical reality is explicit, consented, provenance-preserved, and clinically validated before the LLM treats the picture as evidence.

Experimental mathematics supplies a complementary proof-status analogy. PSLQ and Euler-sum searches can identify striking candidate relations, but a discovered relation is not the same object as a proof, and an uncertain memory of a source is not the same object as provenance. Medical LLM literature mining has the same risk: a plausible relation, even one discovered by a powerful search procedure, must remain labeled as hypothesis or surrogate until proof, validation, or patient-outcome evidence exists.

The noncommutative AM-GM counterexample probe makes this point more concrete. Recht and Re's optimization motivation, Zhang's four-matrix inequality, and the later Lai-Lim disproof of the broader Recht-Re conjecture sit in a literature where simulations can support claims that later fail under exact adversarial construction. In this package I include only a bounded exact-arithmetic check: a 2 by 2 rank-one projector example violates the candidate bound by the factor $1 + \sqrt{2}$. The clinical analogy is not about matrices; it is that medical LLM evaluation should search for small, exact failure modes instead of trusting smooth aggregate simulations.

Sequential depletion ordering adds a runtime-budget analogy. If each safety action consumes a fraction of remaining clinician attention, audit time, compute, or provenance bandwidth, then the order of actions is not neutral. In a single-resource synthetic experiment, small-first order minimizes raw residual-fraction cost, while large-first order minimizes convex log-depletion risk. In heterogeneous multi-resource settings, the locally preferred order can reverse with the residual state. This is why the capstone treats FIFO as an audit-replay policy, not as a universal safety policy.

Harchol-Balter's queueing-theory and scheduling work is the systems version of this point. A hospital-facing agentic LLM is a queueing system under load: it schedules documentation, provenance checks, clinician interruptions, model calls, and escalation paths. Heavy-tailed workloads and resource-allocation policies can change response time and safety behavior, so runtime assurance must monitor scheduling policy rather than assuming that all checks can happen eventually with the same effect.

Consumer heart-rate guidance supplies the simplest wearable example. Harvard Health notes that resting heart rate varies across people and can be affected by stress, anxiety, hormones, medication, physical activity level, age, and fitness. A wearable stream is therefore useful monitoring evidence, but not a standalone diagnostic or exercise-prescription authority. The runtime monitor should preserve that distinction before a medical LLM converts a number into reassurance, alarm, or advice.

3. Methods

I implemented a deterministic synthetic stress-test harness with 30 scenarios derived from course assignments, instructor feedback, and the cited literature. Each scenario specifies the requested claim, endpoint type, study design, citation status, population fit, context, opportunity cost, clinician authority, and evidence-chain status. The final stress cases treat recurring hospital genetics result triage as support-only; treat multilingual, cross-script, and picture-language artifacts as audit-only until source, language, encoding, simulation map, and clinical context are established; treat a real but nonclinical speculative topological theory citation as a proof-status/provenance stress case rather than medical evidence; treat Kolmogorov-Arnold Network architecture claims as model-selection evidence rather than clinical permission; treat fixed FIFO/LIFO runtime processing as unsafe when scarce budgets may be depleted in an order-sensitive way; route residual-alpha claims to orthogonal-projection confounding audit; and preserve family genomic consent boundaries when ranked preferences cycle or privacy budgets are exceeded. The monitor returns one of several runtime actions: allow, allow bounded training simulation, narrow, deny, abstain, stress-test, stress-test ordering, stress-test confounding, preserve consent boundary, or escalate to workflow diagnosis.

For multimodal audio and telemedicine, the core runtime-assurance pattern is deliberately simple: the model may infer, but output is gated before playback or presentation.

This can be implemented in an async or coroutine style: inference produces an intermediate object, execution suspends for an assurance check, and only a permitted representation is resumed into the user-facing output stream.

Liquid neural networks are included only as a candidate implementation family for future streaming monitors. Their continuous-time dynamics and small recurrent state make them a natural architecture to test

for ECG, audio, wearable, or telemedicine streams, but the assurance layer remains model-agnostic: liquid, transformer, classical, and rules-based models all have to pass the same endpoint, provenance, and authority gates before output is shown.

```
async def process_audio(chunk):  
    tracks = await separator.infer(chunk)  
    safe_output = await assurance_controller.check(tracks)  
    await audio_output.play(safe_output)
```

I then added a ZDD-style sparse-family audit. Each LLM response is represented as a subset of evidence features, such as verified citation, endpoint type, study design, population fit, context status, opportunity-cost status, authority boundary, and evidence-chain status. Zero-suppressed decision diagrams are appropriate because most evidence features are absent in any given claim. The audit tests whether observed stress states and permitted hard-outcome or surrogate claim forms can be represented compactly without enumerating the full power set.

Finally, I implemented a model-selection layer. The monitor compares candidate claim models and minimizes an information-criterion-style loss that penalizes missing evidence, forbidden boundary crossings, model complexity, nested dependency burden, and authority risk. This makes the safe output a selected claim form, not merely a refusal template: the monitor can choose second-reader assistance, surrogate-only language, confirmatory-surrogate language, hard-outcome summary, stress testing, or abstention.

The KAN stress case is included here because it is a clean model-selection example. A critical architecture review may legitimately inform when smooth basis-function models are plausible, when benchmark failures diagnose data mismatch, or when computational overhead matters. But those are method claims. They do not become medical outcome, deployment, or safety claims until the clinical evidence dimensions are also present.

This is the agentic core of the proposal. The model may search, parse, compare, and propose, but the hidden safety function is evaluated by the runtime controller. The setup is analogous to AI Safety Gridworlds: the observed reward of producing a helpful answer is separated from the performance function that measures whether the answer respected safety, evidence, and authority boundaries.

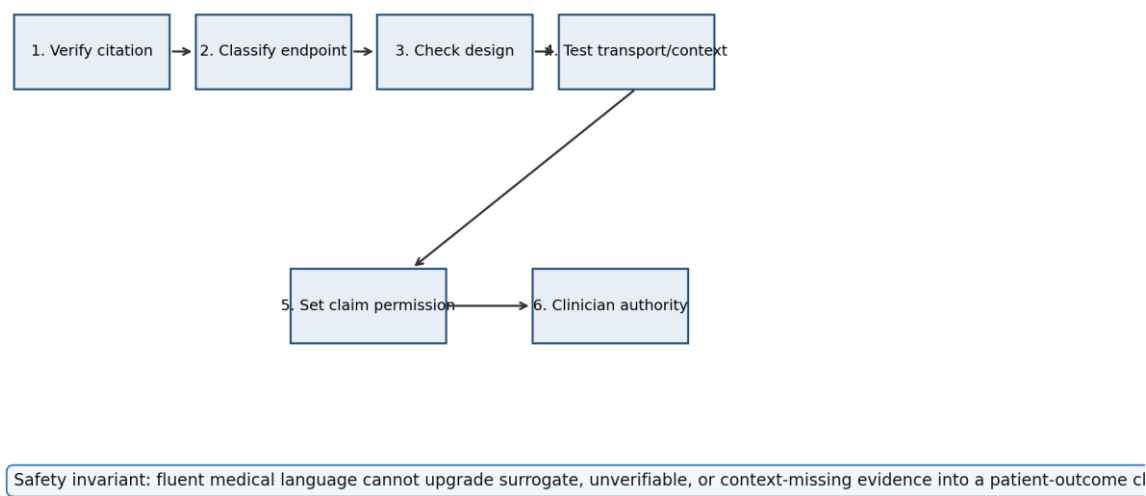


Figure 1. Runtime assurance architecture for evidence-gated medical LLM claims.

Gate	Inputs	Runtime rule
Citation	Verified, local audit, course artifact, unverifiable	Unverifiable citations are denied.
Endpoint	Hard outcome, validated surrogate, surrogate, process, local correction	Only hard outcomes can support hard-outcome claims.
Design	Pragmatic RCT, RCT, validation, observational, case	Study design caps the permission level.
Transport/context	Population match, mismatch, temporal shift, missing material context	Mismatch triggers abstention or stress testing.
Provenance	Raw transcript, cross-script text, encoding state, source language	Unverified artifacts remain audit-only.
Opportunity cost	Addressed, unknown, unaddressed high	High unaddressed cost blocks deployment claims.
Authority	Clinician retained, public-health boundary needed, handoff missing	The model never becomes the final decision-maker.

4. Results

The harness matched all expected actions: 30 of 30 scenarios. Only the hard patient-outcome pragmatic RCT scenario received permission for a cautious clinician-facing outcome summary. The synthetic telemedicine and robotic-cadaver rehearsal cases were allowed as training simulation only. The surrogate detection-rate, recurring hospital genetics triage, KAN architecture, audio/captioning benchmark, SIK3/tinnitus, saliva

evolutionary-genomics, consumer heart-rate/wearable physiology, and Kanazawa-style bounded evidence ambiguity cases were narrowed. The altitude/circadian and reverse-Flynn norm-drift cases were routed to generalization stress testing. The sequential-depletion case was routed to runtime ordering stress testing; the residual-alpha case was routed to confounding/projection audit; and the family genomic-consent case preserved consent boundaries rather than aggregating ranked preferences into one family authorization. The Connes/Penrose evidence-patch case abstained from transport rather than quotienting away context. The cross-script transcript, speculative topological-theory, and picture-language simulation cases abstained for provenance/proof-status audit. The fabricated-citation case was denied, the Yablo-style evidence-chain case abstained, and the CLOT case escalated to workflow diagnosis rather than blaming clinicians for low uptake.

The depletion-ordering experiment enumerated 24 permutations of four synthetic runtime loads with reserve 3.0. It found the expected ordering split: increasing load order minimized raw residual-fraction cost (1.2035), decreasing load order maximized it (1.4589), while decreasing load order minimized convex log-depletion risk (0.9133). In the heterogeneous two-resource probe, the same pair had positive swap delta in one tail state and negative swap delta in another, demonstrating context reversal.

The orthogonal-projection experiment reduced the maximum absolute design-matrix inner product from 38.378 before projection to 4.68e-15 after projection. This supports the confounding gate: a residual signal should not be interpreted as independent medical evidence until its relationship to known design covariates has been audited.

The Kelly-style exposure experiment reproduced Ferguson's favorable-game warning in abstract form: all-in play had high expected value but a ruin probability of 1.000 over 20 rounds. The monitor therefore caps claim exposure at 0.25 while reliability is still uncertain.

The uniform-witness experiment enumerated 180 possible endpoint/design/population/provenance witnesses and found 29 admissible witnesses. A biased strongest-five display had variation distance 0.828 from the uniform admissible witness distribution, showing how curated examples can make evidence look stronger than the underlying claim family.

The family genomic-consent experiment detected a Condorcet-style cycle over 3 consent options across 3 relatives. Full raw genome reuse was blocked, trait-specific partial inclusion remained both consent-feasible and privacy-budget-feasible, and the composed epsilon for partial-plus-audit use was 0.85 under a synthetic epsilon budget of 1.00. This is not a HIPAA revision by itself; it is a runtime test showing why family-linked genomic records need partial permissions and budget accounting.

The finite event-algebra experiment built a sample space of 144 runtime histories and treated unsafe promotion, privacy exceedance, consent collapse, and audit trigger as measurable events. The union and complement identities held to numerical error below 0.0e+00, making the planning rule explicit: histories in the audit-trigger event are routed to narrowed or audit-only output.

The measure-on-measures experiment defined 4 synthetic source-population measures and a meta-measure mixture over them. The maximum total-variation distance from an individual source distribution to the mixture was 0.106; the source with highest audit-trigger probability was `genomeindia_like`. This supports source-specific testing rather than relying only on aggregate performance.

The Mahalanobis covariate experiment compared 4 synthetic source-population vectors over 6 covariates. It flagged 2 source(s) for source-specific validation at threshold 2.0; the largest distance was 2.408 for genomeindia_like. This turns population/covariate fit into an auditable transport gate.

The exact matrix counterexample probe verified a minimal 2 by 2 failure mode for a simulation-supported noncommutative AM-GM-style inequality. The rank-one projector case has right-hand side $1/4$ and left-hand side $(1 + \sqrt{2}) / 4$, giving violation factor $1 + \sqrt{2}$. The positive-definite perturbation with epsilon $1/10$ still violates the bound with margin $3\sqrt{2}/10 - 198/625$. In the capstone this is used only as an evaluation-design warning: reassuring simulations should be paired with small adversarial witnesses.

The tail/maximal-inequality probe enumerated 4096 synthetic 12-step mean-zero runtime-drift paths. With threshold 6 and total variance 12, Kolmogorov's bound was $1/3$; the exact crossing probability was $189/1024$ and the terminal-only exceedance probability was $299/2048$. Because some paths crossed the boundary and returned inside by the terminal time, the probe supports pathwise runtime monitoring rather than final-output-only review.

The Hex boundary-invariant probe enumerated 66066 full boards across sizes 1, 2, 3, 4. It found 0 both-crossing boards and 0 neither-crossing boards, so every full board had exactly one crossing. A small minimax check also found first-player wins for sizes 1, 2, 3. The majority-triangle subcheck enumerated 8 local triple patterns with 0 ties. In the capstone this is a no-ambiguous-terminal-state and safe-coarse-graining check for runtime actions, not a clinical result.

A larger Hex scaling probe then sampled 36000 full boards over sizes 5, 7, 9, 11, 13, 15. The sampled terminal ambiguity count remained 0, and the unbiased blue-crossing mean was 0.503. However, a generic same-size local majority smoother flipped the global crossing in up to 0.256 of sampled boards, with the maximum at side length 15 and blue probability 0.5. This is the operational lesson for genetic privacy blurring and multimodal summarization: aggregation can look locally reasonable while changing the safety boundary.

The ZDD-style audit used 39 possible evidence features. The 30 scenarios collapsed to 23 distinct sparse feature subsets, represented by 89 ZDD nodes versus a naive trie upper bound of 899. The hard-outcome family admitted the pragmatic RCT scenario and rejected surrogate cases; the surrogate/process-support family admitted surrogate-overclaim, recurring hospital genetics triage, KAN architecture, audio/captioning, SIK3/tinnitus, saliva evolutionary-genomics, and Kanazawa bounded-ambiguity cases; the bounded-training family admitted telemedicine and robotic rehearsal cases; the provenance-gap family admitted the cross-script transcript audit case, the speculative topological-theory proof-status case, and the picture-language simulation case; the orthogonal-projection family admitted the residual-alpha case; and the consent-boundary family admitted the family genomic-consent case.

The proof-status poset audit enumerated 4096 states over 6 graded evidence dimensions and 18432 cover transitions. Only 4 states permitted a hard-outcome claim, while 0 nonclinical states and 0 endpoint-free states were promoted. NIETTU remained audit-only under all source/proof-only upgrades. This makes the graded-poset point explicit: proof/source status is not exchangeable for endpoint, transport/context, or clinician-authority status.

The picture-language diagram audit classified 9 diagrammatic or mathematical artifacts, including Jaffe-Liu picture-language simulations, an Axelrod-style repeated-game noise case, and a Bourgade-Huang loop-equation universality case. None authorized a hard patient-outcome claim. 5 artifacts reset validation after representation transfer, 1 required explicit virtual-state labeling, and the loop-equation case remained universality-audit-only: approximate invariant hierarchies can support portability tests, not clinical permission by resemblance.

The loop-equation/Gronwall stability probe then made that portability rule executable. It checked 6 required invariant gates and compared a stable multimodal transfer chain with an unstable missing-provenance chain. The stable chain ended at error 0.0173 under budget 0.050; the unstable chain ended at 0.1792 over budget 0.050. A single-entry perturbation calculation stayed below its synthetic budget, and the switching-cumulant subcheck canceled the main quadratic term while keeping rare-event and replacement-error terms inside budget. The permission therefore remained universality-audit-only rather than outcome permission.

The branch-factor path probe added a collision/branch-cut version of the same rule. A separated multimodal branch with $R_0=36.06$ preserved half-planes, positive phase margin, separation, and Volterra contraction ratio 0.0416. A near-collision family/population branch with $R_0=0.078$ failed the separation/contraction audit and was routed to branch-stability abstention. This operationalizes the BBGKY warning: pointwise-looking identities away from collisions do not justify near-collision genomic, population, or contact-term claims.

The model-selection audit compared 19 candidate claim models and matched the expected safety action in 30 of 30 scenarios. This result is intentionally modest: it shows that claim permission, multimodal simulation boundaries, and ordering stress tests can be made explicit before clinical deployment, not that a medical LLM has improved outcomes.

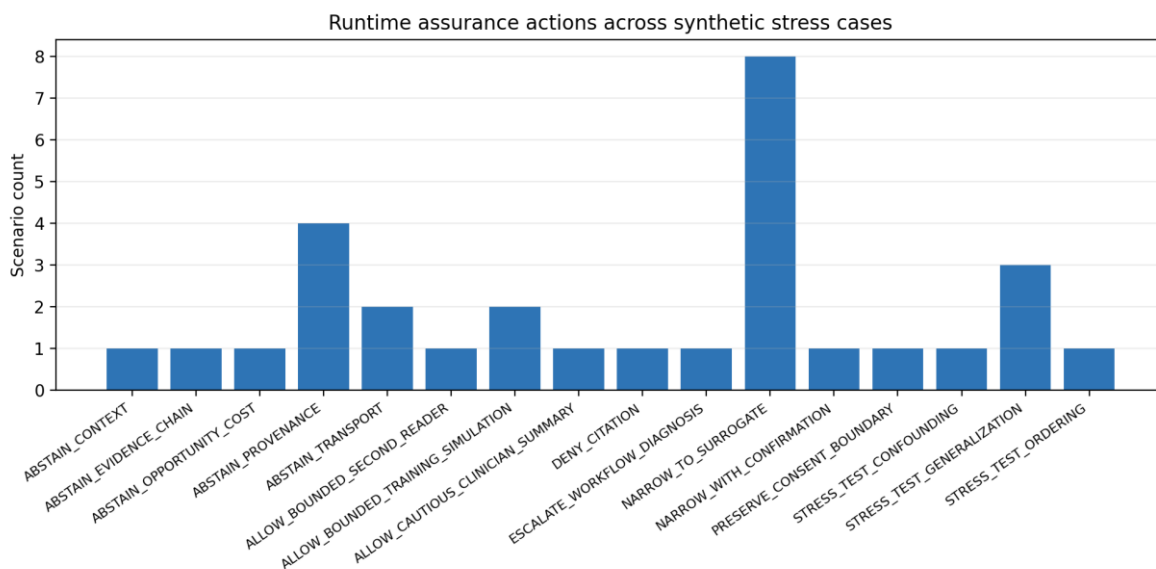


Figure 2. Runtime actions across synthetic stress cases.

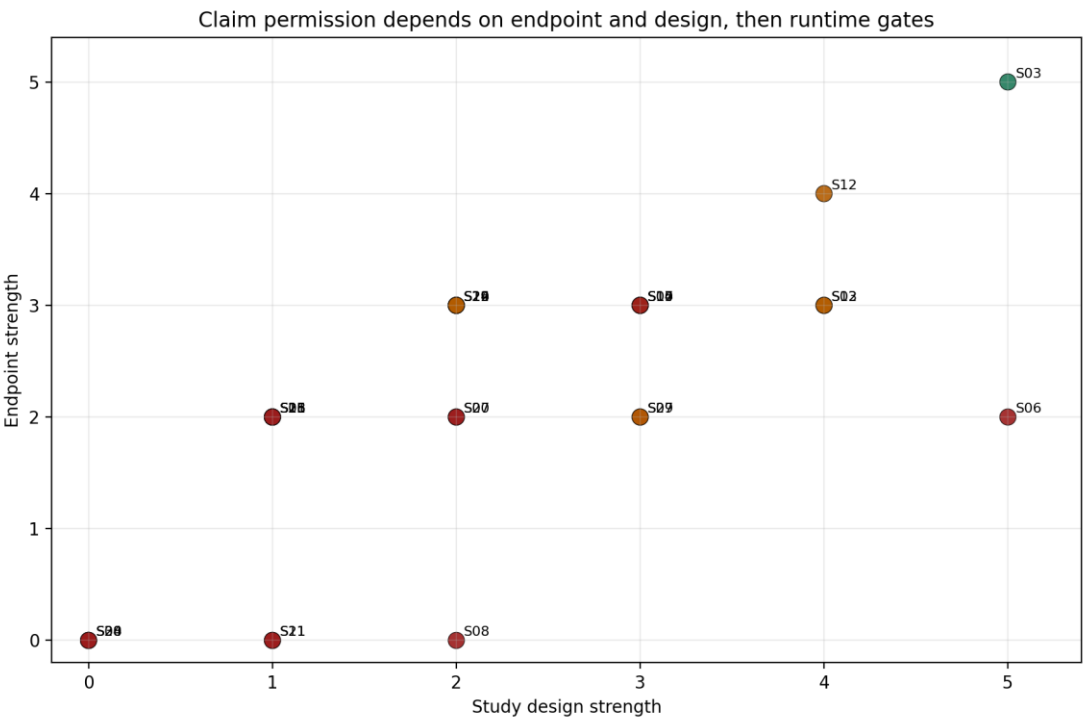


Figure 3. Endpoint and design strength map. Runtime gates still apply after scoring.

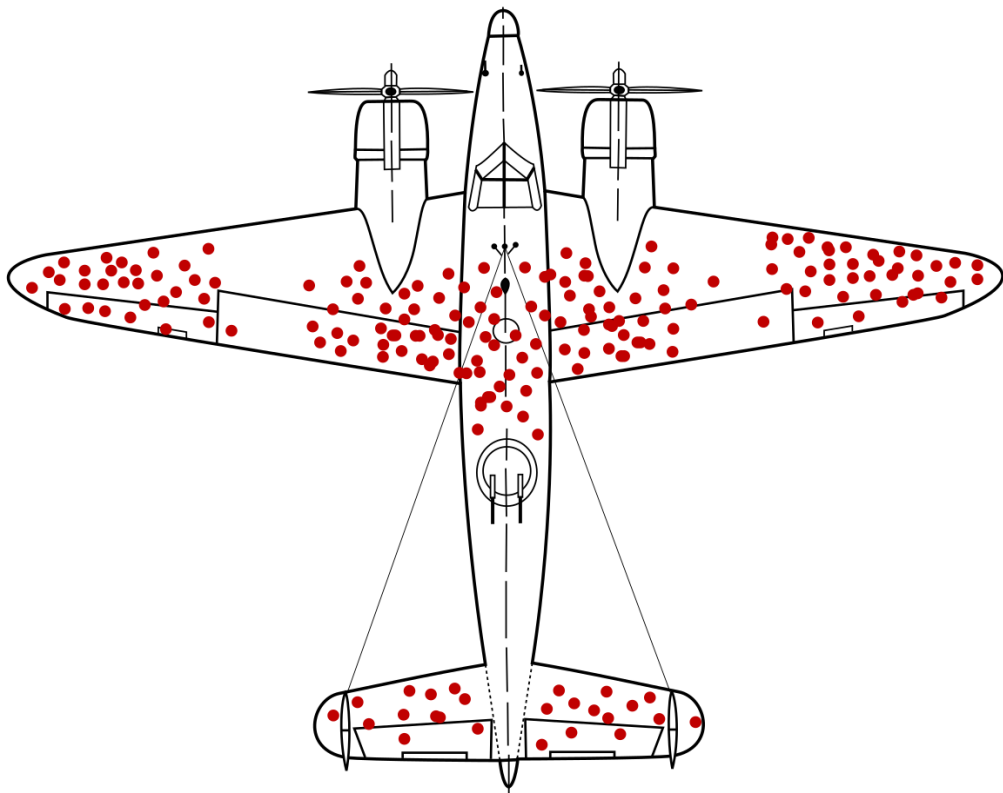


Figure 4. Survivorship-bias analogy for medical LLM evaluation: the missing failures, abstentions, and non-returning evidence chains matter. Image: Grandjean/McGeddon, Wikimedia Commons, CC BY-SA 4.0.

ID	Scenario	Endpoint	Runtime action
S01	MRI second-reader human-baseline correction	local case or error correction	allow second reader
S02	Surrogate detection-rate overclaim	surrogate	narrow surrogate
S03	Hard patient-outcome pragmatic RCT	hard patient outcome	allow outcome summary
S04	Fabricated or unverifiable citation	none	deny citation
S05	Population transport gap	surrogate	abstain transport
S06	CLOT low acceptance	process or workflow	workflow diagnosis
S07	ANA opportunity-cost screen	process or workflow	abstain cost
S08	Yablo-style deferred evidence chain	none	abstain chain
S09	Pinochet context-missing gesture claim	process or workflow	abstain context
S10	SEIR variant generalization shift	surrogate	stress test shift
S11	Consumer genetics feature gap	none	abstain provenance
S12	Validated surrogate with confirmation plan	validated surrogate	confirmatory narrow
S13	Kanazawa-style bounded evidence ambiguity	surrogate	narrow surrogate
S14	Audio separation/captioning metric overclaim	surrogate	narrow surrogate
S15	Telemedicine practice simulation	process or workflow	allow training simulation

ID	Scenario	Endpoint	Runtime action
S16	SIK3 hearing/tinnitus tolerance overclaim	surrogate	narrow surrogate
S17	Altitude/circadian auditory transport overclaim	surrogate	stress test shift
S18	Robotic cadaver/synthetic patient rehearsal	process or workflow	allow training simulation
S19	Reverse-Flynn neuropsych norm drift	surrogate	stress test shift
S20	Quasiperiodic evidence-patch collapse	process or workflow	abstain transport
S21	Cross-script transcript provenance gap	none	abstain provenance
S22	Saliva evolutionary-genomics clinical overclaim	surrogate	narrow surrogate
S23	Sequential depletion runtime-budget ordering	process or workflow	stress test ordering
S24	Consumer heart-rate wearable overclaim	surrogate	narrow surrogate
S25	Orthogonal nuisance component confounding	process or workflow	stress test confounding

S26	Family genomic consent aggregation	process or workflow	preserve consent boundary
S27	Recurring hospital genetics result triage	process or workflow	narrow surrogate
S28	Speculative topological theory provenance boundary	none	abstain provenance
S29	KAN architecture claim model-selection boundary	surrogate	narrow surrogate
S30	Picture-language multimodal simulation boundary	none	abstain provenance

5. Discussion

The result is a small but concrete standard for medical LLM evaluation. A model fails not only when it fabricates. It also fails when it is technically defensible at the wrong evidentiary strength, in the wrong population, at the wrong time, or with the wrong opportunity cost. This is why the capstone treats evidence as an asset and clinical claim strength as a currency. A verified citation is not freely exchangeable into any claim the model can phrase.

For bioinformatics, the same rule applies to privacy and consent. Genetic records are not isolated facts about one patient; they are correlated with relatives, ancestral populations, and future interpretations. Differential privacy gives the right lens because it treats privacy as a measurable loss under repeated analysis, while Arrow's theorem warns against pretending that ranked family preferences always aggregate into one legitimate group decision. The proposed runtime monitor therefore preserves a graded poset of partial release states: full raw genome reuse, trait-specific inclusion, audit-only use, or no reuse may be incomparable across family members and clinical purposes.

The framework also reconciles several course themes. The Module 4 human-baseline example supports bounded second-reader use: an LLM can help catch a possible error without becoming the authority. The Module 5 assignment becomes the central evidence-overclaiming test. The Module 6 Pinochet gesture argument becomes a context gate: an action pattern does not mean the same thing when the material, handoff, or procedural phase changes. The Module 7 SEIR simulation becomes a distribution-shift gate: a day-14 versus day-26 peak shift is exactly the kind of change a monitor should detect before allowing confident predictions.

Yablo's paradox without self-reference supplies a useful warning about citation chains: the chain can be unstable even if no individual source is fabricated. Borrowing only metaphorically from Nakayama-style reasoning, a local citation may support a narrow claim without generating the stronger global claim that an intervention improves patient outcomes in the target population.

Survivorship bias makes the evaluation problem even sharper. A benchmark that records only successful-looking LLM answers is like looking only at the planes that returned. The evaluation log must also preserve refusals, hidden denominator cases, translation failures, dropped citations, missing subgroups, and prompts where the correct action is silence or escalation. Runtime assurance is partly a way of making the non-survivors visible before they become invisible evidence gaps.

The skater/Hawkes replay video included in the package is not medical evidence. It is a visual analogy for runtime assurance: delayed commands, speed envelopes, and separation constraints are monitored separately from the agent's nominal behavior. Medical LLMs need the same separation between fluent generation and safety envelopes.

The reverse-sprinkler result supplies one more implementation analogy. Smith and colleagues show that reversing flow through a sprinkler does not merely invert the ordinary sprinkler mechanism; the geometry of the arms controls momentum flux and therefore rotation. Medical LLM workflows have a similar asymmetry. Generating a fluent clinical answer from evidence and reconstructing the supporting evidence from a fluent answer are not inverse operations unless the architecture preserves provenance, endpoint category, and boundary-crossing history.

Audiology is a useful future stress domain because it makes surrogacy unavoidable. Genetic etiology, caption accuracy, speech perception, word recognition, device performance, and cochlear-implant use can all be clinically meaningful, but none is identical to communication, autonomy, quality of life, or Deaf cultural identity. SIK3 is therefore handled as a guarded hypothesis case: it can support hearing-associated genetic language, and tinnitus GWAS work supports polygenic and neuropsychiatric links, but neither proves tolerance for hearing loss, tinnitus adaptation, counseling benefit, or quality-of-life improvement. The film CODA is a concise cultural reminder of the same point: the relevant outcome is not simply normalization to hearing, but communication access and patient-centered flourishing.

Altitude and circadian rhythm sharpen the transport problem. High-altitude auditory studies show that hypoxia, hypobaria, exposure duration, testing modality, and acclimatization can change measured auditory function. Circadian disruption is likewise a health modifier, while Dinacharya gives a culturally legible vocabulary for daily rhythm and routine. These are context features, not outcome proof. A runtime monitor should therefore stress-test any hearing, tinnitus, sleep, or wearable inference that moves across altitude, routine, ancestry, or care setting.

The positive version of the same idea is telemedicine practice. In a synthetic or standardized-patient environment, audio separation, captions, multimodal replay, and LLM feedback can help clinicians rehearse webside manner, turn-taking, language access, uncertainty disclosure, and escalation boundaries. The runtime permission is training-only: the system may critique the simulated encounter, but it may not imply that the tool improves real patient outcomes until a separate patient-level evaluation supports that claim.

The multilingual transcript case extends this to source provenance. A dub, caption file, OCR scrape, right-to-left Hebrew page, Mongolian Cyrillic devotional text, or mixed-script transcript can be meaningful in its own domain, but it is not automatically a clean medical input. Before an LLM may use such material in clinical reasoning, the runtime layer should verify source, language, encoding, speaker identity, translation status, clinical relevance, and consent boundary. This is the practical reason the paper argues against raw text as the default medical LLM input: text is already a transformation of the encounter.

Toosarvandani's language-memory essay sharpens the healthcare version of this point. For marginalized language communities, speech is not just a carrier of propositional content; it can encode family memory, identity, discrimination, and access to care. Romani adds a concrete technical example because dialect diversity, endangered varieties, and Para-Romani contact varieties can defeat naive language labels. A

runtime monitor for multimodal clinical AI should therefore treat language capture, captioning, translation, and summarization as governed transformations, not as neutral preprocessing steps.

Robotics extends the same boundary. The useful question is not whether robots are 'better than us' in a science-fiction sense, but whether robots, synthetic organs, and humanoid standardized patients can function like cadaver-like rehearsal surfaces: repeatable, inspectable, and safe for mistakes. Literature on synthetic organs and humanoid patient robots supports training and feasibility claims, but the runtime gate should block autonomous treatment claims or patient-benefit claims until separate clinical evidence exists.

The reverse-Flynn neuropsychology example makes the same point for psychometrics. A high-secure psychiatric population showed temporal and population-specific shifts in WAIS full-scale IQ and processing-speed profiles. A medical LLM should not treat general-population norms, old test versions, repeated IQ testing, or ordinary cognitive expectations as portable without checking the population, decade, instrument version, and clinical pathway. This is a runtime transport gate, not a generic intelligence claim.

The Connes/Penrose analogy explains why the project uses sparse relational evidence states rather than only scalar confidence. Classical quotienting says two cases are equivalent when the local patch looks the same; runtime assurance asks which relations, contexts, and permissions survive before a claim is made. This is the same reason the capstone uses ZDD-style sparse families: the safe object is not just an answer, but a membership check in a structured family of admissible evidence states.

The control-theory analogy is likewise bounded. While the claim policy is smooth, it resembles a Hamilton-Jacobi-Bellman problem: choose the safest action under an evidence-state value function. When a claim reaches a permission boundary, that value function becomes nonsmooth and viscosity-solution or distributional language becomes more appropriate. Korteweg-de Vries-style dispersive regularization is useful only as a metaphor for the boundary layer: the dangerous region is the oscillatory transition where surrogate claims begin to travel like outcome claims. The monitor's job is to detect that transition, not to claim that medical evidence literally obeys a wave equation.

6. Limitations

- The scenarios are synthetic and derived from course artifacts and public literature, not from live patient data.
- The harness tests a safety architecture, not clinical effectiveness.
- No medical advice, diagnosis, treatment recommendation, or patient-facing product is produced.
- The next stage requires clinician review, a preregistered rubric, blinded LLM outputs, and eventually pragmatic trial designs if patient-outcome claims are made.

7. Conclusion

Medical LLMs need evidence gates, not just better prompts. The proposed runtime-assurance framework gives a practical path: verify the citation, classify the endpoint, score the design, test population and context, account for opportunity cost, and preserve clinician authority. The most practical first use case is recurring

hospital genetics triage, where verified but bounded artifacts must be routed without being converted into treatment authority. The framework is conservative by design. That conservatism is what lets the system be useful without pretending that polished language is clinical proof.

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